Deterministic Optimization

Robust Optimization

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Concept of Robustness

Robust Optimization

Learning Objectives for this module

- Explore the impact of uncertainty on decision making
- Understand the concept of robustness
- Formulate robust optimization

Example: Drug Production

- A company produces two kinds of drugs, Drug I and Drug II, containing a specific active agent A, which is extracted from two kinds of raw materials, Raw I and Raw II, purchased on the market.
- Production data:
    - Selling prices of Drug I and Drug II ($/1000 packs): 6200, 6900
    - Content of agent A in Drug I and Drug II (g/kg): 0.5, 0.6
    - Man power required (hours/1000 packs): 90.0, 100.0
    - Equipment required (hours/1000 packs): 40.0, 50.0
    - Operational cost ($/1000 packs): 700, 800
    - Purchasing price of Raw I and Raw II ($/kg): 100.00, 199.90
    - Content of agent A in Raw I and Raw II (g/kg): 0.01, 0.02

Example: Drug Production

- Resource constraints:
  - Budget ($): 100,000
  - Man power (hour): 2,000
  - Equipment (hour): 800
  - Capacity of raw material storage (kg): 1,000
- Objective: find the production plan that maximizes the profit of the company
- We can easily formulate a linear programming problem

Example: Drug Production

$$
\begin{array}{lll}
\min & \overbrace{100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII}^{\text{Purchasing and operation cost}} - \overbrace{(6200 \cdot DrugI + 6900 \cdot DrugII)}^{\text{Revenue}} & \\
\text{s.t.} & 0.01 \cdot RawI + 0.02 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0 & \Leftarrow \text{Balance of agent A} \\
 & RawI + RawII \leq 1000 & \Leftarrow \text{Storage constraint} \\
 & 90 \cdot DrugI + 100 \cdot DrugII \leq 2000 & \Leftarrow \text{Manpower constraint} \\
 & 40 \cdot DrugI + 50 \cdot DrugII \leq 800 & \Leftarrow \text{Equipment constraint} \\
 & 100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII \leq 100000 & \Leftarrow \text{Budget constraint} \\
 & RawI, RawII, DrugI, DrugII \geq 0. & \Leftarrow \text{Nonnegativity constraint}
\end{array}
$$

- Optimal solution: RawI = 0, RawII = 438.789, DrugI = 17.552, DrugII = 0
- Optimal objective: -8819.658

Uncertainty in Active Agent

- The contents of active agent in the raw materials cannot be exactly 0.01g/kg for RawI and 0.02g/kg for RawII
- In reality, these contents vary around the indicated values
- Assume for RawI, the active agent varies 0.5% around 0.01, i.e. between 0.00995 and 0.01005, and takes each extreme value with 50% probability
- Assume for RawII, the active agent varies 2% around 0.02, i.e. between 0.0196 and 0.0204, and takes each extreme value with 50% probability
- How does this randomness affect the optimal production plan?

Impact on Production and Profit

- Optimal production:
  - RawI = 0, RawII = 438.789, DrugI = 17.552, DrugII = 0
- With 50% chance, content of active agent in RawII is 0.0196g/kg, this results in $438.789 \cdot 0.0196 = 8.6003$g active agents
- But to produce 17.552 units of DrugI needs $0.5 \cdot 17.552 = 8.776$g active agent
- The optimal production plan becomes infeasible!
- Say in this case the company adapts to the low content of active agent in raw material and cuts production to $\frac{8.6003}{0.5} = 17.201$
- In this case, the profit is $17.201 * (6200 - 700) - 199.9 * 438.789 = 6891.6$

The Unreliability of Nominal Solution

- In the drug company example, a 2% change in the active agent data will cause a 21% drop in profit with 50% chance, and a 11% drop in profit on average!

- The source of the problem is that the company only plans according to the nominal value of the uncertain data, which produces a nominal solution that is very unreliable

- The way to overcome this issue is the concept of <u>robustness</u>

The Concept of Robustness

- We want the production plan to survive “all realizations” of the content of active agents:
    - The solution (RawI, RawII, DrugI, DrugII) should satisfy

$$activeI \cdot RawI + activeII \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$$

for all $activeI \in [0.00995, 0.01005]$, $activeII \in [0.0196, 0.0204]$

- This constraint is called a **Robust Constraint**
- It’s equivalent to a usual deterministic constraint using “worst case”:

$$0.00995 \cdot RawI + 0.0196 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$$

Robust Drug Production Problem

- Incorporating the robust constraint, we have a robust drug production problem

$$
\begin{aligned}
\min \quad & 100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII - (6200 \cdot DrugI + 6900 \cdot DrugII) \\
\text{s.t.} \quad & 0.00995 \cdot RawI + 0.00196 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0 \\
& RawI + RawII \leq 1000 \\
& 90 \cdot DrugI + 100 \cdot DrugII \leq 2000 \\
& 40 \cdot DrugI + 50 \cdot DrugII \leq 800 \\
& 100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII \leq 100000 \\
& RawI, RawII, DrugI, DrugII \geq 0.
\end{aligned}
$$

- Robust optimal solution: RawI = 877.732, RawII = 0, DrugI = 17.467, Drug II = 0, OPT = -8294.567

Summary

- We learned:
  - The issue with only planning for the nominal case: uncertainty in data